

AutoJourn: Framework for News article generation using Prompt Engineering on LLMs and Evaluation mechanisms

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Abstract

*In this paper we present **AutoJourn**, a complete framework for generating news article leveraging open-source LLMs and state-of-the-art prompt engineering techniques. Overall, our pipeline works as follows: extraction of social media discussion, dynamic topic modeling, topic aware summaries and news articles generation. To evaluate our pipeline we used both quantitative (e.g., NPMI) and qualitative (human and AI-based) techniques. The quantitative methods highlight how our LLM-based approach outperforms traditional models (e.g., LDA and BERTopic for dynamic topic modeling), especially the Ollama Mistral. On the other hand, the qualitative evaluation from both humans and AI-based models ensures readability, consistency, and factuality in the generated news articles. Our work consists of an important step in the direction of automated journalism, the choice of technologies was based on open source options to facilitate and make future research accessible to all.*

1 Introduction

The landscape of journalism has undergone significant transformations with the rising popularity of social media and non-traditional media, for which, people are rapidly turning to social media platforms for information and breaking news, often prioritizing them over traditional media sources. The emergence of generative Artificial Intelligence (AI) offers the potential to use the vast amount of information and discussions on these platforms, serving both as an assistant to journalists and as an independent content creator. Recent research indicates that social media’s instant nature has significantly shortened the news cycle. This has created pressure for traditional media outlets to deliver faster reporting while still maintaining comprehensive coverage (Romero). The integration of generative AI into journalism is exemplified by initiatives such as The

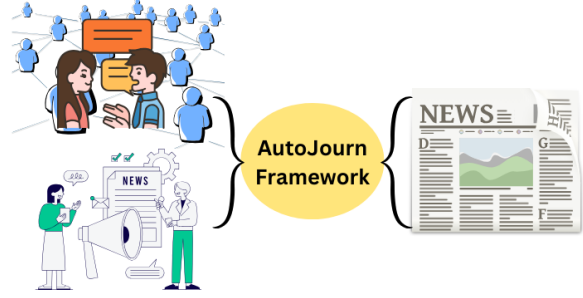


Figure 1: AutoJourn Framework

New York Times adopting AI tools in its newsroom, including for editing, summarizing, coding, and writing (Verge). The Guardian Media Group has formed a partnership with OpenAI to incorporate its journalism content into ChatGPT, making it available to users worldwide. This collaboration represents a strategic decision to adopt AI technology for both creating and distributing content (Guardian). However, the use of AI in journalism also raises concerns about accuracy and credibility. Situations where AI-generated content has led to errors, impose the need for human oversight and clear guidelines in AI-assisted reporting (Notebook).

In this work, we propose an Automated Journalist Application that leverages social media discussions to generate news articles. Our application operates in several stages: (1) It collects the social media conversations based on user search-query using platform APIs, (2) Performs dynamic topic modeling, (3) Generates topic-aware summaries of the extracted discussions, (4) Generates a news article given a chosen topic, the corresponding summary, and an appropriate tone.

Our contributions in this paper are as follows:

- We present the framework behind **AutoJourn**: our Automated Journalist Application, with the aim to produce high-quality, informative, and coherent news articles grounded to the

content.

- We discussed a **Reverse Prompt Engineering strategy** to formulate proper prompts for news generation.
- We present a **Qualitative Evaluation Process** based on a dataset created from our Application.
- We present a new evaluation approach to fact-check and **detect hallucinations** in the generated news articles.

2 Related Work

Here we present a thorough literature review with respect to the several stages of our application pipeline, to identify recent technologies that align better with the specific steps we have to implement.

Dynamic Topic Modeling (DTM)

DTM has been the subject of much recent research, as it overcomes the limitations of the static version. Early approaches relied on probabilistic models; however the advent of Large Language Models (LLMs) greatly improved interpretability, adaptability, and coherence of the DTM tasks.

The early approaches to topic modeling are presented in (Vayansky and Kumar, 2020) and one of the most famous models is Latent Dirichlet Allocation (LDA), a probabilistic model which treats documents as mixtures of topics and topics as distributions over words.

LLMs found applications in various research domains, e.g. DTM. In (Eklund et al., 2022) they proposed a dynamic extension of Neural Topic Modeling by Clustering Embeddings (NTM-CE), which leverages pre-trained language models (PLMs) for document embeddings and applies distance-based clustering to extract topics and track topics evolution over time.

Recently, prompt based approaches were proposed to overcome the limitations of traditional models like LDA and BERTopic. The most relevant approaches are (Wang et al., 2023), (Mu et al., 2024), and (Pham et al., 2024), which leverages LLMs capabilities by employing prompt-based techniques (zero-shot or few-shot prompting) to improve topic coherence, diversity, and evolving topics over time.

Summarization

Throughout the years, summarization has been the subject of many researches, which recently led

to advanced methods with higher summarization quality through prompt engineering, topic modeling, and deep learning approaches.

For prompt engineering, two approaches are (Li et al., 2023) and (Zhang et al., 2022). The first one generates task-specific prompts based on a smaller policy model, it enhances the LLMs performances by dynamically guiding them and reducing the risk of generating off-topic or incoherent summaries. Similarly, (Zhang et al., 2022) enhances BART-based abstractive summarization by introducing contrastive learning and non-negative log-likelihood (NLL) loss to align controlled and uncontrolled summarization modes.

For topic-aware summarization, (Benedetto et al., 2024) is based on a two-step process to generate topic-aware summaries of social media posts: (1) Topic detection, clustering using BERTopic, and (2) Abstractive summarization, using PRIMERA. Another approach is (Wang et al., 2024), an extractive summarization approach to enhance the coherence of summaries by integrating sentence-level topic representations.

News generation

Once topics are extracted and summaries are generated, we use this as a source to generate a well-structured and coherent news article.

Early approaches like StatsMonkey, Monok and Heliograf were used for automated journalism with structured data. However, given the unstructured nature of social media content, these approaches are not feasible for our purpose.

Another approach for news generation are neural-based models. In (Sym et al., 2022) such models are categorized into template-based, pipeline-based, and fully neural architectures.

To improve coherence in long-form generated text, (Chen et al., 2020) proposed a multitask approach, combining end-to-end generation with a decomposed approach. Text coherence is further improved with (Tang et al., 2022), by using contextualized prompts and continuous inverse prompting.

(Hua, 2020) introduced a two-step neural model to perform style-controlled text generation, first with content selection through extraction of key-phrases and then with assignments of roles, to ensure logical flow to sentences. Another improvement was brought by (Ji and Huang, 2021), which enhances long-form text generation by improving discourse coherence through discrete latent variables and step-wise decoding control.

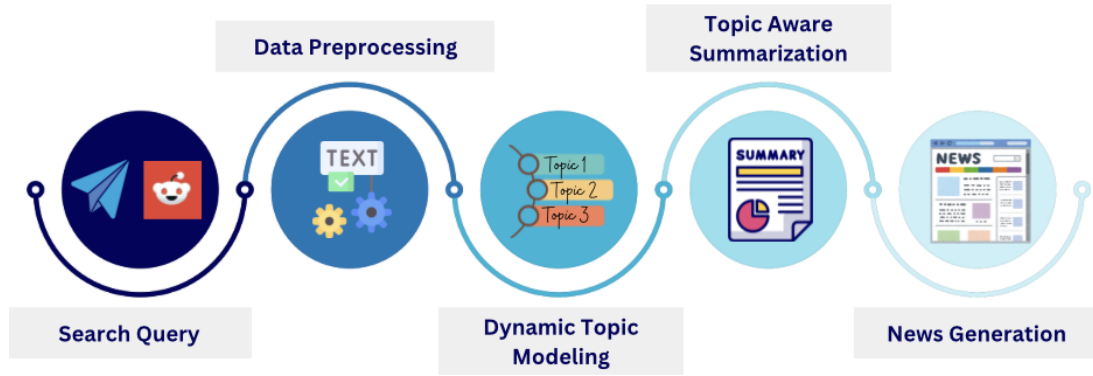


Figure 2: Application Pipeline

In the end, we explored prompt engineering techniques into (Sahoo et al., 2024) where case studies were provided to show how journalists leverage structured prompt to get high quality and coherent long-from generated text.

Generated Article Evaluation

The evaluation of generated articles is developed based on the individual works of (Thorne et al., 2018) where they introduced a large-scale benchmark for retrieving evidence (primarily from Wikipedia) and determining entailment or contradiction for a given claim. The work of (Pang and Lee, 2004) demonstrated the significance of distinguishing subjective expressions from factual statements which basically underpins the modern pipeline to skip fact checking the subjective/creative expressions. The MNLI corpus (Williams et al., 2018) provided a broad-coverage resource for natural language inference which became crucial to textual entailment checks. Techniques such as BERT (Devlin et al., 2018) and BART (Lewis et al., 2019) further popularized large-scale pre-training and finetuning leading to both text classification and NLI-based fact verification.

3 Methodology

3.1 Model Selection and Setup

We employed several pre-trained Large Language Models (LLMs) using both Ollama and Hugging Face frameworks, including Mistral, Phi, Qwen, Gemma and Llama for zero-shot topic modeling. We followed these steps for model setup. Ollama was installed and initiated via the command-line interface. The pre-trained models were fetched from their respective repository. The models were interfaced using LangChain’s OllamaLLM module, enabling structured prompt-based inference.

After going through the experiments described in Section 4, we found Mistral¹ to be the optimal choice for our application pipeline. Mistral is a 7B parameter model which outperforms Llama 2 13B on all benchmarks and Llama 1 34B on many benchmarks according to Ollama.

3.2 Dynamic Topic Modeling

We designed a prompt-based pipeline for topic extraction, allowing the LLM to infer relevant topics along with percentage distributions and keyword sets. The pipeline employs a structured prompt based on zero shot prompting mechanism, instructing the model to analyze a given document and extract the top five most relevant topics along with their respective percentages and 15 associated keywords. Then the processed text was fed into the Ollama Mistral model via LangChain’s invocation mechanism. The raw LLM output was extracted and parsed into structured JSON format using regex-based extraction. The Appendix A shows the zero-shot prompting used for this purpose. The Fig. 3 shows the pie-chart visualisation for the extracted topics based on their likelihood percentages, for a certain conversation in our application.

3.3 Topic-aware Summarization

In our system, after the initial pipeline extracts topics, percentages, and keywords from the original text, this information is utilized in a subsequent stage dedicated to topic-aware summarization. In this stage, the extracted data is fed into a prompt (see Appendix A) instructing the language model to generate concise summaries tailored explicitly to the given topic and its associated keywords. The generated summaries are then stored in the Redis

¹<https://ollama.com/library/mistral>



Figure 3: Pie-chart example for Topic Modeling

client’s back-end, allowing the webpage to dynamically render the relevant summaries when a user clicks on a corresponding topic. The design of this summarization prompt draws inspiration from the framework proposed by Li et al. (Li et al., 2023), which introduces the concept of directional stimulus prompting. We adapt this idea to guide the model’s output and ensure it generates concise summaries centered on the keywords and topic. We integrate this prompt into our pipeline to ensure the generated summary is as close to the user’s selected topic as possible. By using this approach, enhancing the user experience by delivering on-demand has been our intention, as well as topic-specific summaries and utilizing advanced prompt engineering techniques to improve system performance.

3.4 News Generation

After the summaries are generated, the user takes the lead in the news generation process. They select a specific topic and writing style, and the system then aggregates the original text, the extracted topics, and corresponding summaries, integrating these with the user’s designated writing style into a single composite prompt. By using this prompt, a key part of our process, we provide detailed guidance that helps the language model write news articles while keeping the important standards of professional journalism.

3.4.1 Composite Prompt Structure

The composite prompt is designed to be very clear and straightforward. It begins by assigning the model a specific role as a professional journalist. We put key details in the prompt, like what the article is about and a short summary, to keep the news

on the selected topic. We also added a list of important words to help spread the main ideas throughout the article, which makes it easier for readers to follow. A dynamic style instruction, selected from a predefined mapping of stylistic categories (e.g., formal, academic, narrative, or satirical), further refines the prompt by guaranteeing that the generated text consistently adheres to the desired tone.

3.4.2 Reverse Prompt Engineering Strategy

A central innovation of our approach is the application of reverse prompt engineering. Unlike conventional methods that design prompts to achieve a desired output, our reverse prompt engineering strategy inverts the process. We pass exemplar texts to the Mistral language model and query what input prompt would have produced such output. This inquiry enables us to deduce the implicit stylistic instructions embodied in the exemplar. Through the systematic execution of this procedure with varying types of datasets—including formal news articles, academic texts, narrative reports, and more—we establish a comprehensive mapping between observed stylistic features and their corresponding prompt components. The insights gained from this process are then compiled into our predefined style dictionary, allowing for control over the style of generated news content.

A designed composite prompt combined with reverse prompt engineering, our methodology improves the interpretability of the news generation system. Ongoing refinements, guided by qualitative and quantitative evaluations, ensure the system consistently produces detailed, contextually rich, and stylistically diverse articles. The prompt detailed in Appendix A is a product of these developments and is further informed by the approaches described in the work of (Khan et al., 2023) on generating text in arbitrary writing styles.

3.5 Whole Pipeline

All these stand-alone modules of prompting LLMs for specific purposes, taken together into the application backend, serves as the whole pipeline behind the Automated Journalist Application. Fig. 2 provides an overview of the pipeline’s workflow.

3.6 Evaluation Metrics

3.6.1 Quantitative Evaluation for Topic Modeling

We evaluated the topic modeling with the following quantitative NLP metrics:

Topic Coherence Score: gives a quantitative measure of the interpretability of a topic (Mimno et al., 2011). It is measured through **Normalized Pointwise Mutual Information (NPMI)**, a score that computes the topic coherence based on word co-occurrence. Proposed by (Lau et al., 2014)

$$\text{NPMI}(w_i, w_j) = -\frac{\text{PMI}(w_i, w_j)}{\log P(w_i, w_j)} \quad (1)$$

where **PMI (Pointwise Mutual Information)** is given by (Jurafsky and Martin, 2025):

$$\text{PMI}(w_i, w_j) = \log \frac{P(w_i, w_j)}{P(w_i)P(w_j)} \quad (2)$$

Topic Diversity Score: (Dieng et al., 2020) defines it to be the percentage of unique words in the top N words of all topics and it ranges from 0 to 1 where a score close to 0 indicates redundant topics and close to 1 indicates more diverse topics.

3.6.2 Qualitative Evaluation for Summarization

Based on the work of (Nguyen et al., 2024), we propose a qualitative approach to evaluate the Summarization and News generation pipeline by employing both automated scores and LLM-based approaches. For this purpose, we create our own dataset consisting of conversations we extracted from our application and their corresponding summary and news. Automated Journalist Dataset ² consists of 5 columns: query, text, topic, summary, news generated from our application. We have used these automatic metrics to evaluate the generated Summary: BERTScore to evaluate similarity, FRE Score to measure readability, DCR Score to check distributional consistency.

We analyzed the results of these metrics and identified the documents where we have low confidence, and put them to the following two approaches: **Human Judgment** where we ask specific subject matter experts to judge the quality of the summary with regards to Readability, Redundancy, Coverage with a Likert scale and **AI-powered Judgment** where we give the relevant text and summary in an advanced engineered Prompt to some of the most advanced LLMs like: ChatGPT 4o, DeepSeek, Perplexity etc. and ask them to do the same as human judgment on the likert scale. Once we get the results for those identified documents, we aggregate the results to deal

²<https://www.kaggle.com/datasets/himelghosh/automated-journalist-dataset/data>

away with the subjectivity difference as multiple human-judgments can have both Intra-personal and Inter-personal variations which we would like to minimise in order to have more confidence through this Qualitative evaluation pipeline. See Fig. 4 for our proposed pipeline. We formulated the prompt for evaluation by several LLMs based on (Zheng et al., 2023) and for LLM hallucination checking we refer to (Hu et al., 2024).

3.6.3 Qualitative Evaluation for Generated News

We follow a two-way approach to evaluate the News article generated. First, we assess the Readability, Coherence and Quality of the article. For this we use classical tools like **Grammarly** ³. Second, we assess the factual correctness with respect to the source text and check for hallucinations.

We employed a new technique for fact-checking (Thorne et al., 2018) and hallucination detection. See Fig. 5 for the illustration. We take the generated article and do sentence segmentation on it. Then we use a subjectivity classifier (Pang and Lee, 2004) to decide whether each sentence is objective (factual) or subjective (creative/opinion based). Using transfer learning process (Devlin et al., 2018), we took a pre-trained DistilBERT classifier and fine tuned it with a large subjectivity-objectivity labelled dataset ⁴ from Hugging Face library. To improve performance we even further fine-tuned with our small labelled dataset relating to journalistic scenario. Then we used this classifier on each sentences of our news dataset.

$$\mathbf{h}_1, \dots, \mathbf{h}_n = \text{TransformerEncoder}(\mathbf{e}_1, \dots, \mathbf{e}_n), \quad (3)$$

where $\mathbf{e}_i$ is the embedding of token x_i , and $\mathbf{h}_i$ are the contextual representations. Often the special token's representation $\mathbf{h}_{\text{CLS}}$ is used as a summary of the sequence. For binary classification (0/1), we project the hidden state of the [CLS] token onto a 2-dimensional space:

$$\mathbf{z} = \mathbf{W} \mathbf{h}_{\text{CLS}} + \mathbf{b}, \quad \mathbf{z} \in \mathbb{R}^2. \quad (4)$$

Then apply the softmax:

$$P(\text{label} = c \mid \mathbf{x}) = \frac{\exp(z_c)}{\sum_{j=1}^2 \exp(z_j)}, \quad c \in \{0, 1\}. \quad (5)$$

The cross-entropy loss to train the classifier is:

³<https://app.grammarly.com/>

⁴<https://huggingface.co/datasets/SetFit/subj>

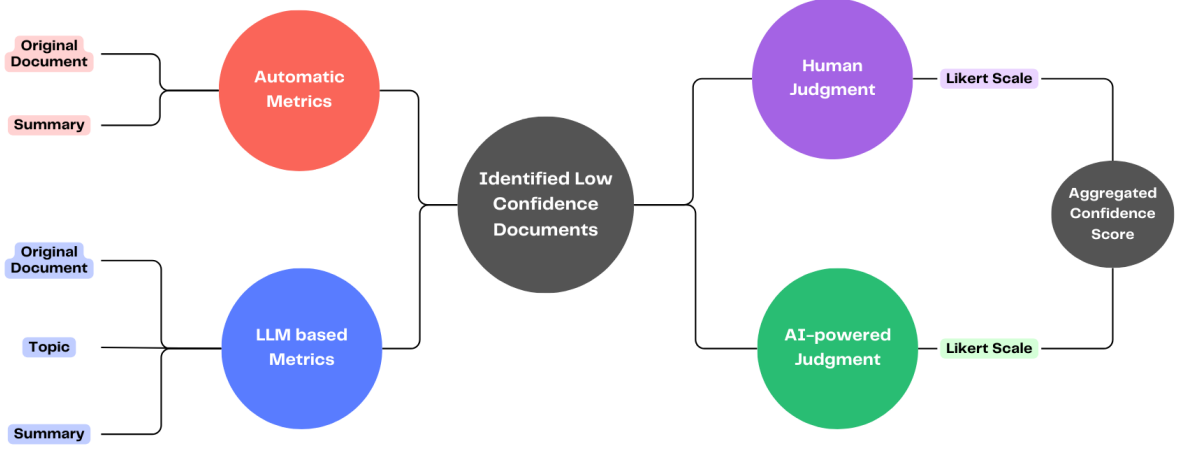


Figure 4: Mind Map for evaluating generated Summary

$$\mathcal{L}_{CE} = - \sum_{c=0}^1 y_c \log[P(\text{label} = c \mid \mathbf{x})], \quad (6)$$

where y_c is a one-hot indicator for the true label.

Afterwards, we incorporated a 3-way fallback mechanism for hallucination detection.

Stage A: We extract snippet from source document and feed it as "premise" and the segmented sentences from generated articles (claims) as "hypothesis" to a textual entailment model (e.g. facebook/bart-large-mnli) (Williams et al., 2018)(Lewis et al., 2019).

Given a premise $\mathbf{p}$ and hypothesis $\mathbf{h}$, the model outputs logits $\mathbf{z} \in \mathbb{R}^3$ for {contradiction, neutral, entailment}:

$$P(\text{label} = c \mid \mathbf{p}, \mathbf{h}) = \frac{\exp(z_c)}{\sum_{j=1}^3 \exp(z_j)},$$

$$c \in \{\text{contradiction, neutral, entailment}\}. \quad (7)$$

We often look specifically at

$$P(\text{entailment} \mid \mathbf{p}, \mathbf{h}) = \frac{\exp(z_{\text{entail}})}{\sum_{j=1}^3 \exp(z_j)}, \quad (8)$$

and compare it to a threshold τ .

If the entailment score is below threshold, the claim is a potential hallucination. Then we fall back to **Stage B**: the pipeline then queries Wikipedia using Named Entity Recogniser (NER) to check for evidence supporting the claim and use the entailment model to return us a probability again. If the probability is below threshold, the claim remains flagged as hallucination and we fall back to **Stage**

C: where we use Google custom search with the entities and extract snippets from search results and re-run entailment. If the score is still below threshold, the final verdict is that the claim is a hallucination. If in any of these stages, the entailment score goes above threshold, the claim is not a hallucination.

If $P(\text{entailment}) < \tau$ with respect to a snippet from the source document, we query Wikipedia, and if still below τ , we query Google. Let $\mathbf{d}$ be a retrieved snippet, and $\mathbf{h}$ the claim:

$$P(\text{entailment} \mid \mathbf{d}, \mathbf{h}) = \frac{\exp(z_{\text{entail}})}{\sum_{j=1}^3 \exp(z_j)}. \quad (9)$$

If none of the stages yields an entailment above τ , we label the claim as a hallucination.

Finally we retrieve the hallucination scores as a percentage. And then we use a parallel pipeline to put the source text and generated article in a well designed prompt to advanced LLMs and ask them to judge on the hallucinations by searching the web because we don't want false positives in hallucination scores. False positives can be possible from our code because of paraphrased or summarised texts in the generated articles. We have given the News evaluation prompts for the LLMs in the Appendix D.

4 Experiments

4.1 Experimental Setup and Datasets

Experiments were conducted on Google Colab with T4 GPU. We used the 20newsgroups dataset (Lang, 1995), a widely used text corpus containing ar-

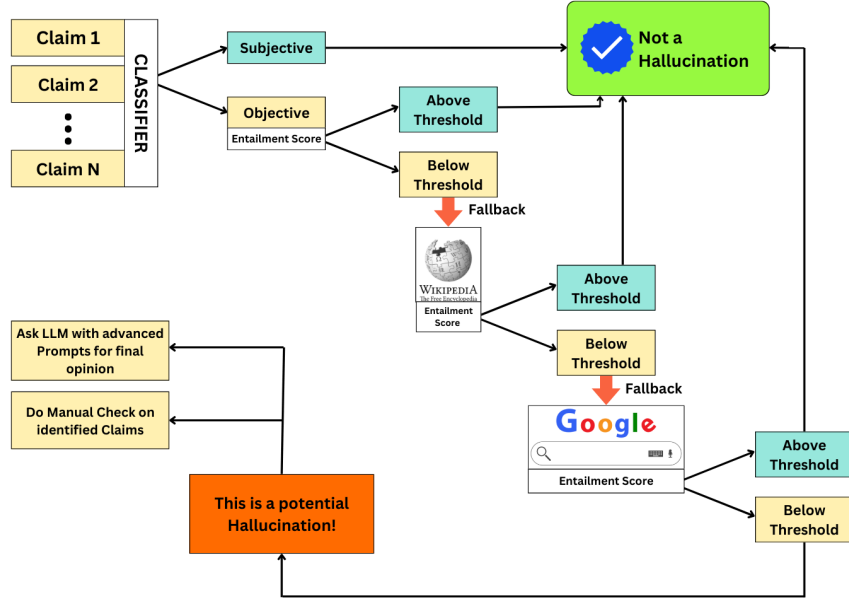


Figure 5: Illustration to describe our hallucination detection strategy

ticles from 20 different categories, in the above-mentioned pipeline to generate our results. To verify generalizability of our results, we also used the BBC News dataset⁵ which comes from (Greene and Cunningham, 2006). See Table 1 for the description.

Dataset	Size	Labels
20 NewsGroup	16,309	20
BBC News	2225	5

Table 1: Dataset Description. Size means the Number of documents. Labels being the number of news categories.

4.2 Results

4.2.1 Quantitative Evaluation Results on Dynamic Topic Modeling

To contextualize our results, we reference Table III from (Wang et al., 2023), which provides NPMI and TD scores for various topic modeling methods applied to the 20 Newsgroups dataset. Consult Table 2 for the NPMI and TD scores of Topic Modeling calculated across our two datasets. The results demonstrate that LLM based Topic Modeling achieves competitive performance with more coherent and diverse topics comparable or sometimes better than BERTopic while LDA and NMF may

struggle with accomodating the semantic meaning of the topics. For 20NewsGroups, the NPMI remained 0.36 and for BBC News, it was 0.68 while the TD for 20NewsGroups being 0.82 and for BBC News being 0.86 exhibits high levels of cohesion and diversity making the ollama mistral model a better choice for this task.

graphicx

Model	20 NewsGroup		BBC News	
	NPMI	TD	NPMI	TD
LDA	-0.05	0.81	0.02	0.88
NMF	0.04	0.63	0.14	0.98
BERTopic	-0.16	0.75	0.16	0.85
Ollama Mistral	0.36	0.82	0.68	0.86

Table 2: Performance comparison of Topic Modeling on 20 NewsGroup and BBC News datasets.

To evaluate the whole pipeline of Summary and News Generation, we employed some qualitative evaluation procedures.

4.2.2 Qualitative Evaluation Results on Summarization

On our application dataset with 10 distinct data points relating to various topics, we found the following Automatic Metric scores in Table 3 for the documents in our dataset.

A BERTscore between 0.75 to 0.85 suggests that the summary retains the semantic meaning of the

⁵<https://www.kaggle.com/datasets/himelghosh/bbc-news-dataset-2226/data>

Topic	BERTScore	FRE Score	Density	Coverage	Redundancy
President Trump Decisions	0.799	40.58	0.582	0.269	0.3818
Ukraine War	0.804	33.44	0.549	1.119	0.549
Political Analysis - Ukraine Conflict	0.807	37.94	0.532	0.278	0.367
Applying to Law Schools (LLM)	0.799	47.18	0.569	0.303	0.463
Diamond Market Scarcity and Pricing	0.813	15.51	0.597	0.120	0.368
Bob Dylan Personality	0.809	44.44	0.702	0.202	0.427
Movie Review: Timothee Chalamet Career	0.792	27.83	0.427	0.365	0.372
Vincent van Gogh	0.765	47.62	0.669	0.242	0.388
Economy and Business	0.796	35.10	0.468	0.298	0.468
COVID-19 Testing Issues in the US	0.821	16.52	0.596	0.114	0.266
COVID-19 Recovery Stories	0.821	33.54	0.649	0.181	0.336

Table 3: *Summary Evaluation Scores on Automatic Metrics calculated by python coding*

original text moderately. FREscore between 20 to 40 suggests that the summary is readable for an educated audience, and above 40 means it is readable by wider audience. Density between 0.4 to 0.6 suggest that the summary compresses information efficiently while avoiding excessive repetition whereas above 0.6 reflect a repetitive use of words from original text. The coverage below 0.3 suggest that the summary doesn't cover most information from the text and a coverage between 0.3-0.7 reflects that it preserves some of the information and above 0.7 it preserves most of the original text information. Redundancy score is expected to be as low as possible showing that summary has minimal repetition of information. Table 3 suggests "COVID-19 Testing" summary has low coverage and low readability and possibly needing more details. "Ukraine war" summary has high coverage and high redundancy, it may not need more information and is already too long. While "Diamond Market" summary with low FREScore is unreadable, and covers less information from the original text. After running these low confidence documents in the LLM-based evaluation pipeline based on engineered Prompt (see Appendix: A) we get their Automatic scores again to re-evaluate. For "COVID-19 Testing", the LLM-based scores ^{6 7 8} are in Table 4.

We have taken an average of these results to pose a concrete idea about the scores and it suggests that "Covid 19 Testing" summary has sufficient coverage and good readability.

Afterwards we use this dataset in the Human and AI-based Evaluation process where we ask a few

individuals and LLMs to judge on the summary based on the conversation. See Appendix B for the questions. We used several LLMs like GPT-4o, Claude, Perplexity, DeepSeek-V3 to judge on the same prompt in order to account for subjective variability of judgments and aggregated the results in an average. After running the judgments through all the documents and LLMs, we found the overall average score to be approximately 2.14/5 for the mistral model generated summaries. While

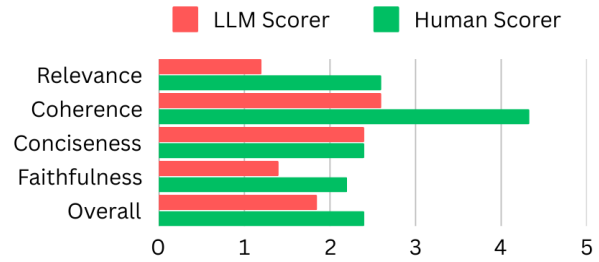


Figure 6: *This Chart illustrates the score given by the LLMs and Human Evaluators on the summary. The data on Vincent van Gogh has been considered for this result.*

for some documents, it performed relatively better, it showed feeble performance in other documents imposing a risk of unrelatedness, hallucinations and divergence from the main conversation. However, it is noteworthy to mention that DeepSeek-V3 did not give any judgments for Political or controversial documents, hence we decided to remove DeepSeek-V3 from the Evaluation process. To be more confident about the results, we take an example case like: "Vincent van Gogh" document from our dataset and do a thorough human evaluation where we ask the same questions as we did with the LLMs and get the ratings which we compare with the LLMs. Here are the responses received: [Human feedback](#). See the Figure 6 for the results. Notably, the results are very close to the LLM based results

⁶<https://chatgpt.com/share/67bcd473-0d0c-8008-a6bc-de019731c755>

⁷<https://claude.ai/share/47c5ab50-989f-42c1-ac5a-b2eb23d610ab>

⁸<https://www.perplexity.ai/search/paste-txt-rZ2vI2qDSx2Iv0JV.riWOA>

Metric	GPT 4o-mini	DeepSeek V3	Claude 3.7	Perplexity AI	Average
BERTscore	0.30	0.65	0.65	0.85	0.61
FREScore	60	60	40	60	55
Density	0.20	0.45	0.50	0.70	0.46
Coverage	0.30	0.50	0.40	0.75	0.48
Redundancy	0.10	0.10	0.70	0.15	0.26

Table 4: Summary Evaluation Scores on Automatic Metrics calculated by LLMs

with an overall average score of 2.4/5.

4.2.3 Qualitative Evaluation Results on Generated News Article

Here we present the results of our novel evaluation pipeline on the generated News articles. All the code and results are available in our [Github Repository](#). We found astounding results on the documents like "Vincent van Gogh" with 0% hallucination detected and "Bob Dylan" with 13% hallucination marked. For the other documents, on a

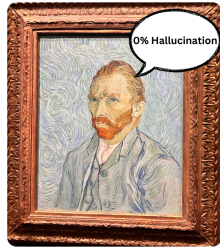


Figure 7: This figure illustrates the results on Generated News on Vincent Van Gogh. The picture was clicked in Musée d’Orsay, Paris by Himel Ghosh in Jan’2025.

manual check, some of the detected hallucinations were false positives mostly flagged from simplified conclusive texts. For this reason, we put this in the LLM to make a final check. The results from Chat GPT-4o are in this [link](#). For all the claims which were falsely marked as Hallucinations in our evaluation technique and we detected them manually to be false positives, are now unmarked by GPT-4o with credible sources and reasons because of its advanced knowledge base, semantic understanding and open access to the web sources. See Table 5 for the results.

The article on Vincent van Gogh remains the best performing with least to no hallucinations after this dual analysis while for most other articles, the average hallucination has been fairly minimal (upto 20 %) from this application pipeline. See Appendix

Document Name	Hallucination %-age	
	Ours	LLM
Trump	31.58	16.67
Ukraine War	25	21.42
Application to LLM	46.15	21.42
Diamonds	66.67	13.34
Bob Dylan	13.64	13.34
Timothée Chalamet	20	33.34
Van Gogh	0	6.67
Economy	23	13.34
Covid Testing	25	6.67
Covid Recoveries	37	53.34

Table 5: Results on Hallucination percentage on generated articles by our detection pipeline and LLM judgment

D for the prompt used.

5 Limitations and Future Works

Although we used free of cost, open-source LLM for the framework, the quality of results are not exactly guaranteed to be comparable to that of a paid premium version of the leading LLMs in the market. Yet we can use this framework to do our work and then evaluate them and suggest further enhancements. The qualitative evaluation is still in many ways, human-dependent and a complete automated approach without errors and without loss of semantic understanding is not available. The automated technique we presented in the news article evaluation section, lacks a fixed ground truth, because the comparison is done on given source-text and the web as a whole. Even then, the process doesn’t fully understand paraphrased or concluded texts and gives false positives. That’s why we kept an LLM based evaluation layer on top of it, for a final judgment. We still have many open avenues to enhance our naive evaluation technique. As a future possibility, we can follow a retrieval-augmented-generation (**RAG**) based approach in our frame-

work for the news article generation where we query more documents from the web and news-sources, use the top-k documents in our prompt for the news generation along with the source text and summary.

6 Conclusion

In this paper, we introduced **AutoJourn**, a complete framework for automating news article generation from social network discussions, by leveraging state-of-the-art prompt engineering techniques on open source Large Language Models (LLMs). Our fully automated pipeline includes: dynamic topic modeling, topic-aware summarization and produces news articles based on a given topic and chosen style.

To validate the goodness of our application we carried out quantitative and qualitative evaluation. We showed how open-source LLM based models can achieve better performance than traditional methods like LDA and BERTopic, with a better topic coherence and topic diversity under the dynamic purview without supervised set of pre-defined topics.

We used open-source LLMs like Ollama Mistral because it does not cost money unlike other LLMs like GPT, DeepSeek, Claude etc. By comparing between some of the free LLMs: Mistral, Qwen, Phi, we saw Mistral outperformed them and decided to use Mistral in our entire pipeline.

We presented **Qualitative Evaluation Approaches** for both summarisation and news generation to verify and judge the performance of the LLM. We took Human judgment and advanced LLM (viz. GPT 4o, DeepSeek R1, Claude) judgments into account to judge the quality of generated summaries. We introduced a new and unique methodology to **fact check** and **find hallucinations** in the generated articles by using subjectivity/objectivity classification using downstream task finetuning on a pretrained DistilBERT model and a three-way fallback mechanism on entailment based fact-checking leveraging source text, wikipedia and google; we also increased confidence on the results by using a layer of LLM-based judgments using GPT-4o. We found the generated news articles by our application framework performed fairly well and with further enhancements, they can be made more accurate. We discussed some of those enhancement possibilities in Limitations and Future Works of our paper.

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7 APPENDIX

A Automated Journalist Prompts

These are the prompts used for the application in the whole pipeline.

Prompt 1 Dynamic Topic Modeling

Require: `preprocessed_doc`, `num_topics`

Ensure: JSON-formatted output containing topics, percentages, and keywords

1: **Prompt:**

Analyze the following text and provide at least `{self.num_topics}` most suitable topics, along with their percentages and the relevant 15 keywords.

Text:

`{preprocessed_doc}`

Format the output as a JSON object:

```
{
  "topics": [
    {
      "name": "Topic1",
      "percentage": 25.0,
      "keywords": ["keyword1", "keyword2", ...]
    },
    ...
  ]
}
```

Prompt 2 Summarization

Require: `topic_name`, `keywords`, `text`

Ensure: Topic-aware summary

1: **Prompt:**

Summarize the following text in 2-3 sentences, focusing on the provided topic and keywords.

Hint: Topic - `{topic_name}`; Keywords - `{keywords}`.

Ensure the summary is concise and captures the main points of the text without adding many details.

If the text is very short, condense the summary into a single, clear, and well-structured sentence.

Avoid repeating keywords unnecessarily or adding unrelated details.

Text: `{text}`

Prompt 3 News Generation

Require: topic_name, keywords, text, summary, tone_instructions

Ensure: Topic-aware summary

1: **Prompt:**

You are a professional journalist writing for a major news outlet.
Your goal is to craft a compelling and detailed news article.

Topic: {topic_name}

Summary: {summary}

Key Points and Keywords: {keywords}

Style: {tone_instructions}

Requirements:

Write a long, engaging news article.

Include an attention-grabbing headline at the beginning.

Expand upon the provided summary using the listed keywords.

Use them naturally throughout the article.

Include historical context, background, or analysis where relevant.

Use the specified style and tone throughout the article.

Instructions:

Start your response with the headline, followed by the full article.

B Human Evaluation Questions for Summaries

You are an expert in text summarization. Your task is to assess the quality of the following summary based on the source text and topic. Provide a score (1 to 5) for each of the following criteria:

1. **Relevance:** Does the summary capture the most important information from the source text?
2. **Coherence:** Is the summary logically structured and easy to understand?
3. **Conciseness:** Is the summary concise without omitting critical details?
4. **Faithfulness:** Does the summary accurately reflect the source content without introducing inaccuracies or hallucinations?

Source Text: *Insert source text here*

Topic: *Insert Topic Here*

Summary: *Insert summary here*

Provide a score for each criterion and a final overall score (average of the four scores)

C LLM Evaluation Questions for Summaries

Prompt 4 Summaries Evaluation

Require: topic_name, text, summary

Ensure: Topic-aware summary

1: **Prompt:**

You are an expert in evaluating text summaries.
Your task is to evaluate the following Summary based on the Source Text and Topic.
Strictly consider the following Automatic Metrics: BERTScore, FREScore, Density, Coverage, Redundancy.

Source Text: {text}

Topic: {topic}

Summary: {summary}

Provide the score for each mentioned metric, in a table.

D Hallucination Evaluation Prompt for Generated News

Prompt 5 Generated News Evaluation

Require: source_text, generated_news

Ensure: Results in a tabular format: number of claims, number of hallucinations

1: **Prompt:**

You are an expert in Hallucination Detection in LLM generated texts and fact checking on the web.
Extract the number of claims in the following News Article and the number of hallucinations out of it.
Detect the creative, subjective sentences and spare them.
For factual sentences, refer the following Source Text, Web, Wikipedia, Google Search for fact checking the claims.
Also keep in mind, some claims are paraphrases or summarisations from the source text and they might not be hallucinations.

Source Text: source_text

Generated text: generated_news
